

AI-Powered WhatsApp Customer Service Proposal

Hello,

My name is Fabricio Santana. I am a software engineer and technical lead with 20+ years of experience in backend development, software architecture, database integration, automated testing, cloud computing, and delivery of critical systems. I have also led large engineering teams and worked on projects where data consistency, business rules, validation, security, and maintainability are essential.

Your project is a strong match for my background because the main challenge is not only connecting AI to WhatsApp. The important part is designing a reliable customer-service workflow where natural-language queries, intent recognition, approved answer generation, escalation to human agents, and operational monitoring work together safely in production.

My relevant approach for this kind of project is to combine backend API integration, AI application design, answer-source preparation, workflow automation, operational guardrails, deployment support, and automated testing into one coherent delivery plan. This keeps the first version practical and bounded while preserving a solid technical base for future expansion.

Working methodology

My delivery method is designed to reduce risk before development starts. The work begins with a fixed Discovery Sprint and then moves into a delivery package only after scope, priorities, responsibilities, dependencies, and acceptance criteria are clear.

The first step is a one-week Discovery Sprint with a fixed price of USD 500. In this phase, I can hold up to five remote meetings to understand the business problem, review the support workflow, analyze the available content and data sources, map requirements, identify risks, review integrations, and define priorities. The deliverable is a practical work plan for the implementation, including recommended scope, technical approach, milestones, success criteria, and the most appropriate delivery package.

After discovery, the project can move into one of three fixed-price, four-week delivery packages. The package is selected according to scope, complexity, delivery speed, integration needs, quality requirements, and the amount of engineering support required. Some roles may participate part-time depending on the agreed scope, especially Scrum Master, technical leadership, test automation, DevOps, observability, and deployment support.

Delivery package options

- **Essential Delivery** — USD 2,000, 4 weeks. Includes a Scrum Master and one full-stack engineer. This package is designed for a focused module, MVP, backend feature, import flow, or small integration with limited uncertainty. It is suitable when the scope is well defined, the number of integrations is limited, and the main goal is to deliver a reliable working feature without unnecessary overhead.
- **Product Acceleration** — USD 4,000, 4 weeks. Includes a Scrum Master, two full-stack engineers, and one engineer focused on test automation, DevOps, observability, and deployment. This package is recommended when faster delivery is needed or when the scope includes a product, API, integration, internal workflow, multiple features, or more demanding quality needs.
- **Scale & Intelligence** — USD 6,000, 4 weeks. Includes a Technical lead, Scrum Master,

two full-stack engineers, and one engineer focused on test automation, DevOps, observability, and deployment. This package is designed for complex systems, advanced data workflows, automation, auditability, security, observability, broader platform evolution, or stronger technical scale planning.

Role responsibilities are clear: the Scrum Master organizes scope, communication, ceremonies, progress tracking, and delivery alignment; full-stack engineers implement backend, frontend, API, and database changes as needed; the test automation, DevOps, observability, and deployment engineer supports automated tests, CI/CD, monitoring, release quality, and deployment; and the Technical lead, included in the Scale & Intelligence package, owns architecture, technical decisions, code quality, and risk management for more complex projects.

For an AI automation project like this, the implementation setup can include provider integration, message intake, AI workflow orchestration, approved answer-source integration, answer-quality evaluation, escalation logic, logging, deployment, and handoff work within the agreed package scope.

Each delivery cycle is executed with Scrum-inspired practices, frequent alignment, technical review, automated testing when applicable, and a demonstration of what was delivered. The client keeps ownership of the produced code, and the scope is agreed before execution to avoid open-ended work and unclear expectations.

Construction defect warranty. After delivery and acceptance, I include a 30-day warranty for construction defects related to the approved scope. This covers fixes for implementation errors, broken agreed behavior, or defects introduced by the delivered code. It does not cover new features, changes in business rules, new intents or answer content not included in the approved scope, third-party service changes, infrastructure changes outside the project, model behavior changes caused by external providers, or requests that expand the original acceptance criteria.

Opportunity analysis

This project should be treated as a customer-service workflow automation problem, not as a generic chatbot setup. A reliable AI-powered WhatsApp responder needs more than a language model. It needs a compliant provider choice, an approved answer-source strategy, clear intent handling, confidence thresholds, fallback behavior, human escalation, conversation logging, and operational visibility after go-live.

The business value is also important: the system should reduce response time, standardize answers to common questions, reduce repetitive manual work, improve handoff to human agents, and create better visibility into support interactions.

The main technical risks are selecting the right WhatsApp-compliant integration model, choosing the correct first AI approach, working with weak or inconsistent business content, defining maintainable escalation rules, handling privacy constraints in customer messages, keeping the first version bounded, and avoiding expansion into CRM integration, analytics platforms, multilingual support, dashboards, or open-ended assistant behavior before the core workflow is stable. I also recommend avoiding unofficial WhatsApp Web automation because it can create account-ban, reliability, and maintenance risks.

A practical first implementation cycle should start with a bounded AI-assisted workflow: official WhatsApp Business API integration, approved support-content ingestion, intent recognition or retrieval-based answering for common inquiries, a small evaluation set of representative questions and expected answers, explicit escalation rules, operational logs, and the technical foundation required for future expansion. The exact delivery boundary should be confirmed during discovery.

Proposed work plan

I would approach the project in structured phases. The Discovery Sprint is the immediate recommended commitment, and implementation can move directly into a fixed delivery cycle once the provider, AI approach, workflow, and acceptance criteria are confirmed.

1. Discovery, support-workflow definition, and technical design

- Review the current support process, message types, FAQ patterns, escalation rules, support-team workflow, expected message volume, privacy constraints, and target success metrics.
- Define the preferred WhatsApp-compliant provider and the correct integration approach for the business case.
- Review the available answer sources and determine whether the first version should use intent classification, retrieval-based answering, templated responses, or a bounded hybrid model.
- Define a small evaluation set with representative questions, expected answers, escalation cases, and acceptance criteria for the first release.
- Produce a concrete implementation plan with workflow map, recommended architecture, acceptance criteria, responsibilities, dependencies, risks, and package recommendation.

2. Core integration architecture and backend foundation

- Design and implement the integration foundation for incoming messages, provider callbacks, AI orchestration, answer-source access, escalation logic, and conversation logging.
- Define data structures and APIs that preserve traceability and allow future extension.
- Add validation, authorization, auditability, and structured error handling.
- Establish the technical base for deployment, monitoring, and secure environment configuration.

3. AI-assisted reply workflow

- Implement the first version of intent recognition, retrieval, or bounded response generation according to the agreed architecture and approved answer sources.
- Keep the first implementation cycle focused on common inquiries and approved support flows rather than broad open-ended chatbot behavior.
- Structure the reply logic so it can be reviewed, tuned, and expanded safely after go-live.
- Support confidence-based decisions for automatic reply, fallback message, or escalation.

4. Human escalation and operational control

- Implement explicit escalation rules for cases the automation cannot resolve safely.
- Route conversations to the correct human agent or support flow according to the agreed business rules.
- Preserve logs, confidence markers, and decision traceability for escalated interactions.
- Define fallback behavior for provider failures, unsupported intents, low-confidence output, or missing business content.

5. Testing, monitoring, deployment support, and handoff

- Test message handling, AI decisions, answer quality, escalation flows, provider integration, and edge cases.
- Add operational visibility for failed replies, low-confidence patterns, provider issues, and unusual usage behavior.
- Document the architecture, provider setup, workflow limitations, maintenance steps, and evaluation approach.
- Provide deployment support and a handoff session or written notes so your team can operate and improve the system safely.

Estimated timeline

For the immediate next step, my recommendation is a 1-week Discovery Sprint. After discovery, the first implementation cycle can move into a fixed 4-week delivery package. For a bounded AI-assisted WhatsApp support workflow with official API integration, intent handling, escalation, and monitoring, the likely recommendation is one Product Acceleration cycle. If the project needs broader NLP behavior, multilingual support, richer knowledge retrieval, or stronger evaluation and observability, the likely recommendation is one Scale & Intelligence cycle.

A practical schedule would be:

- Week 1: Discovery Sprint, workflow mapping, provider recommendation, answer-source review, AI-approach decision, evaluation set, acceptance criteria, and implementation plan.
- Weeks 2 to 5: first fixed implementation cycle, covering the agreed AI-assisted support workflow and operational handoff.

If discovery shows that the first version is narrower than currently implied, the implementation may fit a simpler delivery shape. If it shows broader AI behavior, more demanding reliability requirements, stronger evaluation needs, or additional integrations, I will recommend the larger package or additional cycles accordingly.

Availability and communication

I can work remotely and keep communication in English or Portuguese through Workana messages, scheduled calls, and written progress updates. I usually prefer short alignment checkpoints during the week, especially after reviewing the workflow and after each major implementation milestone.

This helps reduce misunderstandings and keeps the project aligned with your actual support process, business rules, knowledge sources, and operational constraints.

Budget

Based on the current understanding, my recommended immediate starting budget is USD 500 for the one-week Discovery Sprint. This gives you a concrete plan before committing to the full implementation budget, while keeping a clear path to delivery in the following implementation cycle.

After discovery, the first implementation cycle will likely fit one of these packages:

- **Product Acceleration** at USD 4,000 for 4 weeks if the scope is a bounded AI-assisted support workflow with official WhatsApp Business API integration, approved content sources, escalation, and operational monitoring.
- **Scale & Intelligence** at USD 6,000 for 4 weeks if the scope includes broader NLP behavior, multilingual support, richer knowledge retrieval, stronger evaluation and observability, or more demanding operational controls.

This assumes that the client provides access to an approved WhatsApp Business API provider or BSP, approved support content, representative customer questions, escalation rules, and the deployment or hosting environment. It also assumes the first release focuses on common inquiries and does not include CRM integration, account-specific transactional automation, a full agent console, campaign messaging, or broad multilingual support unless those items are confirmed during discovery.

The Discovery Sprint will refine the scope and confirm the correct implementation package. If the review shows a narrower and tightly bounded workflow, I can recommend a smaller delivery shape. If it shows broader support operations, higher message volume, stricter quality controls, or additional integrations, I will recommend the larger package or additional cycles.

Questions before final confirmation

Before confirming the final scope, I would like to clarify a few points:

1. Which WhatsApp provider or business setup is currently available: Meta official API, Twilio, Zenvia, 360dialog, or another provider?
2. What content should the AI use as its approved answer source: FAQs, help-center articles, SOPs, product documentation, or another source?
3. Should the first version only automate common inquiries, or should it also handle transactional or account-specific questions?
4. Is there already historical support data that can be used for intent mapping, answer examples, or evaluation?
5. Should the first version use intent classification plus approved responses, retrieval-based answering, or a bounded hybrid model?
6. What should trigger escalation to a human agent: low confidence, sensitive topic, unsupported intent, negative sentiment, or another rule?
7. What message volume do you expect per day or per month?
8. Are there language requirements beyond English, and is multilingual support part of the first version?
9. What success metrics matter most: response time, containment rate, escalation accuracy, CSAT, or resolution time?
10. Who will approve access to the WhatsApp provider, knowledge sources, and deployment environment?

Once these points are clear, I can confirm the final delivery plan and milestones.